

COL7001 – Advanced Computer Networks

Assignment 1 – Part 0

Background Research – Real BGP Data

1. BGP Looking Glass Tools Used

Two public BGP monitoring tools were used for this investigation:

1. Hurricane Electric (HE.net) BGP Toolkit / Super Looking Glass
2. RIPEstat Looking Glass

The investigation was performed using the Autonomous System **AS132780** and the IPv4 prefix 103.27.8.0/24.

Both tools were used to observe the BGP route and the ASes appearing in the routing path.

2. Institute ASN and Advertised Prefixes

The HE.net AS information page identifies:

AS132780 – Indian Institute of Technology Delhi

The HE.net Prefixes v4 page shows four IPv4 prefixes associated with AS132780:

- 103.27.8.0/24
- 103.27.9.0/24
- 103.27.10.0/24
- 103.27.11.0/24

The screenshot shows **100% visibility** for each of these prefixes, with **1015/1018** displayed for each route.

For the detailed route investigation, 103.27.8.0/24 was selected.

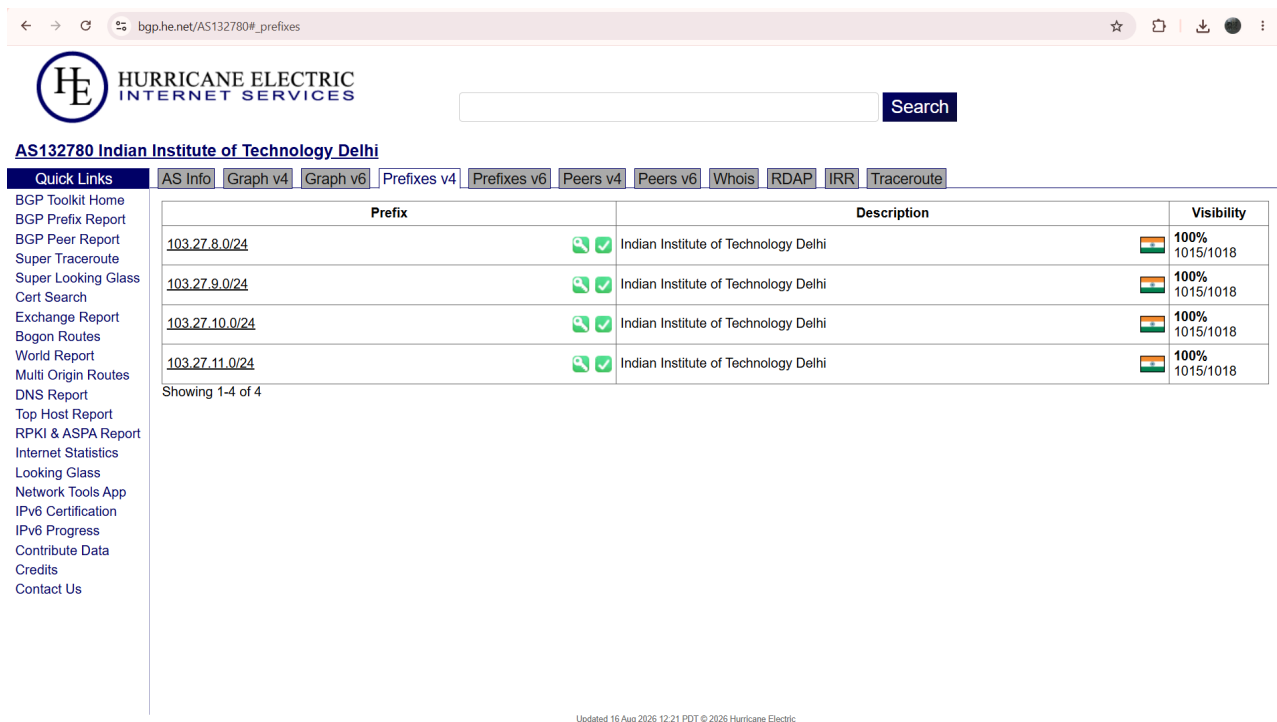


Figure 1: HE.net Prefixes v4 page for AS132780 showing the four advertised IPv4 prefixes and their visibility.

3. HE.net BGP Route Observation

The HE.net Super Looking Glass was queried for the prefix 103.27.8.0/24.

The screenshot shows the following AS path:

20965 -> 9885 -> 55824 -> 132780

Reading the path starting from the IIT Delhi origin gives:

AS132780 -> AS55824 -> AS9885 -> AS20965

The screenshot identifies the immediate neighbor as:

62.40.96.16 (AS20965)

and identifies the route as being learned from **route-views8**.

The route information shown by HE.net also contains the following attributes:

- Origin: **IGP**
- RPKI validation: **VALID**
- MED: **0**
- Communities: 20965:9885, 20965:65530, and 55824:309

The RPKI status displayed in the screenshot is **VALID**, with the message that at least one VRP matches the route prefix.

Quick Links

[BGP Toolkit Home](#)
[BGP Prefix Report](#)
[BGP Peer Report](#)
[Super Traceroute](#)
[Super Looking Glass](#)
[Cert Search](#)
[Exchange Report](#)
[Bogon Routes](#)
[World Report](#)
[Multi Origin Routes](#)
[DNS Report](#)
[Top Host Report](#)
[RPKI & ASPA Report](#)
[Internet Statistics](#)
[Looking Glass](#)
[Network Tools App](#)
[IPv6 Certification](#)
[IPv6 Progress](#)
[Contribute Data](#)
[Credits](#)
[Contact Us](#)

Super Looking Glass

Terminal

103.27.9.0/24

Processing....

✓ Exact Match

+ Shorter

+ Longer

+ Brief

+ Text

+ Exclude

All ASNs

Select a neighbor to query directly ▼

1087 Paths observed

BGP data is delayed up to 5 minutes

Neighbor	62.40.96.16 (AS20965) Learned from: route-views8
Prefix	103.27.9.0/24
AS Path	20965 9885 55824 132780
Origin	IGP
RPKI validation	VALID At least one VRP Matches the Route Prefix
MED	0
Communities	20965:9885 20965:65530 55824:309
Last Updated	8/8/2026, 1:37:42 PM (1w02d08h)
Neighbor	62.40.96.21 (AS20965) Learned from: route-views8
Prefix	103.27.9.0/24
AS Path	20965 9885 55824 132780

Figure 2: HE.net Super Looking Glass observation for 103.27.8.0/24.

4. IIT Delhi BGP Peer

The HE.net Peers v4 page for AS132780 shows one peer in the captured result:

AS55824 – NKN Core Network

Therefore, from the supplied screenshot, AS55824 can be identified as the IIT Delhi peer visible in the HE.net result.

This also agrees with the HE.net Super Looking Glass path, where AS55824 occurs immediately before AS132780 when the path is read from the origin towards the rest of the observed path.



AS132780 Indian Institute of Technology Delhi

Quick Links

BGP Toolkit Home
BGP Prefix Report
BGP Peer Report
Super Traceroute
Super Looking Glass
Cert Search
Exchange Report
Bogon Routes
World Report
Multi Origin Routes
DNS Report
Top Host Report
RPKI & ASPA Report
Internet Statistics
Looking Glass
Network Tools App
IPv6 Certification
IPv6 Progress
Contribute Data
Credits
Contact Us

[AS Info](#) [Graph v4](#) [Graph v6](#) [Prefixes v4](#) [Prefixes v6](#) [Peers v4](#) [Peers v6](#) [Whois](#) [RDAP](#) [IRR](#) [Traceroute](#)

Rank	Description	IPv6	Peer
1	NKN Core Network 	X	AS55824

Updated 16 Aug 2026 12:21 PDT © 2026 Hurricane Electric

Figure 3: HE.net Peers v4 page for AS132780 showing AS55824, identified in the screenshot as NKN Core Network.

5. RIPEstat BGP Route Observation

The RIPEstat Looking Glass was also queried for 103.27.8.0/24.

The screenshot reports:

358 peers announcing 103.27.8.0/24, originated by 132780 and seen by 23 RRCs.

The RIPEstat result shows multiple AS paths for the same prefix.

One observed path is:

513 -> 9885 -> 55824 -> 132780

Another observed path is:

559 -> 20965 -> 9885 -> 55824 -> 132780

When the second path is read starting from the IIT Delhi origin, it becomes:

AS132780 -> AS55824 -> AS9885 -> AS20965 -> AS559

Thus, the screenshot provides an observed route containing four AS hops after the IIT Delhi origin:

1. AS55824
2. AS9885
3. AS20965

4. AS559

Only AS55824 is given an organization name in the supplied screenshots:

The RIPEstat screenshot also shows:

- Next Hop: 192.65.185.3
- Peer: 192.65.185.3
- RRC: **RRC04 (CIXP in Geneva, Switzerland)**
- Last Updated: 2026-08-08T08:07:39Z

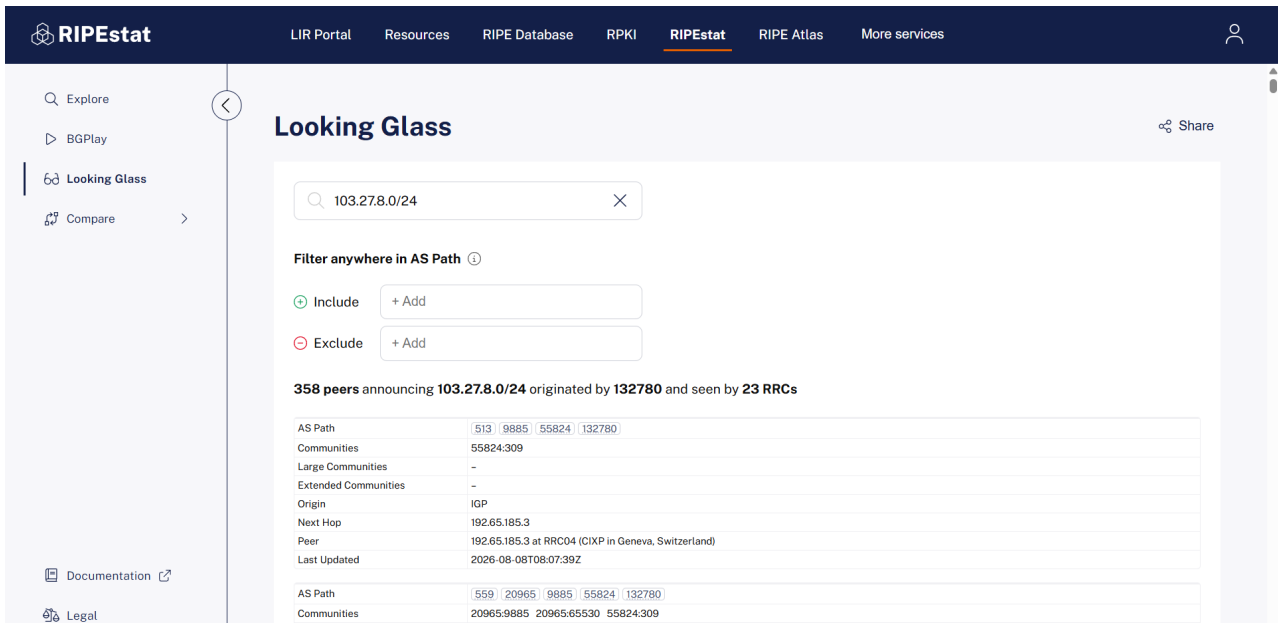


Figure 4: RIPEstat Looking Glass result for 103.27.8.0/24, showing multiple AS paths and observations from 23 RRCs.

6. Comparison of the Two Looking Glasses

The two looking-glass tools do not show exactly the same complete AS path.

The HE.net screenshot shows:

AS132780 -> AS55824 -> AS9885 -> AS20965

The RIPEstat screenshot shows, among other paths:

AS132780 -> AS55824 -> AS9885 -> AS513

and:

AS132780 -> AS55824 -> AS9885 -> AS20965 -> AS559

The common portion visible in the supplied screenshots is:

AS132780 -> AS55824 -> AS9885

The difference occurs farther along the observed AS path.

This discrepancy is expected when comparing BGP observations from different looking-glass systems because the tools can obtain route information from different BGP observation points.

The RIPEstat screenshot explicitly shows that the prefix is observed by **23 RRCs**, while the HE.net screenshot identifies the route as being learned from **route-views8**. Therefore, the two screenshots represent different observations of the same BGP prefix.

The important result from the supplied screenshots is that both tools identify **AS132780** as the origin and show **AS55824** and **AS9885** as part of the route, while RIPEstat exposes additional AS-path variations.

7. Summary

The investigation was performed for IIT Delhi's AS132780 using the IPv4 prefix 103.27.8.0/24. The HE.net Prefixes v4 screenshot shows that AS132780 advertises four IPv4 prefixes: 103.27.8.0/24, 103.27.9.0/24, 103.27.10.0/24, and 103.27.11.0/24. All four prefixes have 100% visibility in the captured result.

The HE.net Super Looking Glass shows the path AS20965 -> AS9885 -> AS55824 -> AS132780, while the RIPEstat screenshot shows multiple paths, including AS559 -> AS20965 -> AS9885 -> AS55824 -> AS132780. The common portion of the observed routes is AS132780 -> AS55824 -> AS9885. The RIPEstat result also reports 358 peers and 23 RRCs for the prefix. Hence, the two tools do not show exactly the same complete AS path, demonstrating the different route observations available from different BGP monitoring points.

Screenshots Included

1. **Figure 1 – HE.net Prefixes:** iitd_prefixes.png
2. **Figure 2 – HE.net Super Looking Glass:** he_looking_glass_103_27_8.png
3. **Figure 3 – HE.net Peers:** iitd_peers.png
4. **Figure 4 – RIPEstat Looking Glass:** ripe_stat_103_27_8.png